24 There are some counters in a bag.
The counters are red or white or blue or yellow.

Bob is going to take at random a counter from the bag.

The table shows each of the probabilities that the counter will be blue or will be yellow.

Colour	red	white	blue	yellow
Probability	0.2	0.1	0.45	0.25

There are 18 blue counters in the bag.

The probability that the counter Bob takes will be red is twice the probability that the counter will be white.

(a) Work out the number of red counters in the bag.

P(red or white) = 1 – 0.45 – 0.25 = 0.30

1 mark
Find the combined unknown probability

R = red; W = white.

R		W		Sum	
2	×0.1	1	×0.1	3	×0.1
0.2		0.1		0.3	

1 mark
Share 0.30 in the ratio 2 : 1

Prob = probability; N = number.

	Prob		N	
	0.45		18	
÷9	0.05	×4	2	÷9
×4	0.2		8	×4

1 mark
Use 0.45 corresponding to 18
to find the number for 0.20

Number of red counters = 8

1 mark
Correct number of red counters

8

(4)

A marble is going to be taken at random from a box of marbles.
The probability that the marble will be silver is 0.5

There must be an even number of marbles in the box.

(b) Explain why.

Exactly half the marbles are silver, and that must be a whole number.

Half an odd total is not whole, so the total must be even.

(1)

1 mark
Use the fact that marbles cannot be split into halves

(Total for Question 24 is 5 marks)